

(Suggested problems)

(20)

Ex(1): A medical researcher wishes to determine if a pill has the undesirable side effect of reducing the blood pressure of the user. The study involves recording the initial blood pressures of 15 college-age women. After they use the pill regularly for six months, their blood pressures are again recorded. The researcher wishes to draw inferences about the effect of the pill on blood pressure from the observations given below. Do the data support the claim that the pill reduces blood pressure? Use $\alpha = 0.05$

Subject	Before (x)	After (y)	d	d ²
1	70	68	2	4
2	80	72	8	64
3	72	62	10	100
4	76	70	6	36
5	76	58	18	(18) ²
6	76	66	10	100
7	72	68	4	16
8	78	52	26	(26) ²
9	82	64	18	(18) ²
10	64	72	-8	64
11	74	74	0	0
12	92	60	32	(32) ²
13	74	74	0	0
14	68	72	-4	16
15	84	74	10	100

$$\sum d_i = 132$$

$$\sum d_i^2 = 2848$$

soln

step(1) $H_0: \mu_d = 0$

$H_1: \mu_d > 0$

step(2) $t_{cal} = \frac{\bar{d}}{s_d/\sqrt{n}}$

$$\bar{d} = \frac{\sum_{i=1}^n d_i}{n} = \frac{132}{15} = 8.8$$

$$s_d^2 = \frac{\sum d_i^2 - n(\bar{d})^2}{n-1} = 120.457$$

$$t = \frac{8.8}{10.975/\sqrt{15}} = 3.11$$

$$s_d = \sqrt{120.457} = 10.975$$

step(3)

Since $t = 3.11 > 1.761$

We reject H_0



$$t_{\alpha, n-1} = t_{0.05, 14} = 1.761$$

Paired - t - test

Ex(2): A study was conducted to investigate the effectiveness of hypotism (الشرع) in reducing pain. Results for randomly selected subjects are shown in the table. The "before" value is matched to an "after" value.

Before	after
6.6	6.8
6.5	2.4
9.0	7.4
10.3	8.5
11.3	8.1
8.1	6.1
6.3	3.4
11.6	2.0

Are the sensory measurements, on average, Lower after hypnotism? Test at a 5% significance level.

Solution: step(1) State the hypothesis

$$H_0: \mu_d \neq 0$$

$$H_1: \mu_d < 0.$$

step(2): calculate the test of statistic.

$$t_{cal} = \frac{\bar{d}}{s_d/\sqrt{n}}$$

d	d^2
0.2	0.04
-4.1	16.81
-1.6	2.56
-1.8	3.24
-3.2	10.24
-2	4
-2.9	8.41
-9.6	92.16
$\Sigma d = -25$	$\Sigma d^2 = 137.46$

$$\bar{d} = \frac{\Sigma d}{n} = \frac{-25}{8} = -3.125$$

$$S_d = \sqrt{\frac{\Sigma d^2 - \frac{(\Sigma d)^2}{n}}{n-1}}$$

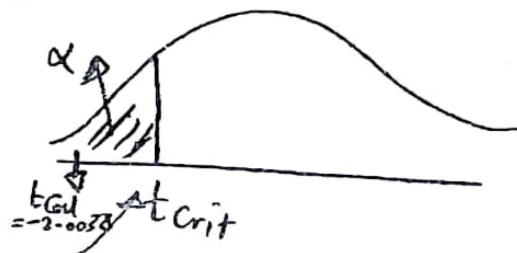
$$S_d = \sqrt{\frac{137.46 - \frac{(-3.125)^2}{8}}{8-1}}$$

$$= 4.411$$

$$t_{cal} = \frac{\bar{d}}{S_d/\sqrt{n}} = \frac{-3.125}{4.411/\sqrt{8}} = -2.0038$$

Step(3): Calculate the critical value

$$t_{crit} = t[\alpha, n-1]$$



$$= t[0.05, 7] = -1.895$$

Step(4): Decision: we reject H_0 .

Thus, Hypnotism appears to be effective in reducing pain.

Dependent paired t-test

Ex (3) A researcher A reading center claims that students will perform better on a standardized reading test after going through the reading course offered by their center. The table shows the reading scores of 6 students before and after the course. At $\alpha = 0.05$, is there enough evidence to conclude that the student's score after the course are better than the scores before the course. one tail

student	1	2	3	4	5	6
Score(Before)	85	96	70	76	81	78
Score(after)	88	85	89	86	92	89

Solution: ^{step(1)} State the hypothesis.

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \geq 0.$$

step(2) : Calculate the test of statistic

$$t_{cal} = \frac{\bar{d}}{S_d / \sqrt{n}}$$

d	-3	11	-19	-10	-11	-11
d ²	9	121	361	100	121	121

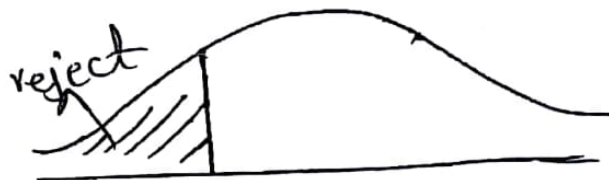
4

$$\bar{d} = \frac{\sum d}{n} = \frac{-43}{6} = -7.167$$

$$S_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n-1}} = 10.245$$

$$t_{cal} = \frac{\bar{d}}{S_d/\sqrt{n}} = \frac{-7.167}{10.245/\sqrt{6}} = -1.714$$

Step(3) ;



Since

$$t = -1.714 > -2.051$$

we accept H_0

$$-t_{\alpha, n-1} = -t_{0.05, 5} = -2.015$$

1) A medical research study was conducted to determine whether there is difference between the effectiveness of two pain-relief medicines used for headaches. Over a 6-month period, a sample of individuals used one of the medicines and a second sample of individuals used the other medicine.

The results is given by.

Test to see whether there is a significant difference between the mean pain-relief times for the two medicines.

Use $\alpha = 0.05$. Assume equal variances.

	Individuals Using medicine (1)	Individuals Using medicine (2)
Sample size	$n_1 = 21$	$n_2 = 21$
Mean	$\bar{x} = 24.8$ minutes	$\bar{y} = 26.1$ minutes
Standard deviation	$s_1 = 3.3$ minutes	$s_2 = 4.2$ minutes

Soln:- step (1): $H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

step (2) $t_{cal} = \frac{\bar{x} - \bar{y}}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = 14.265 \Rightarrow s_p = 3.77$$

$$t = \frac{24.8 - 26.1}{3.77 \sqrt{\frac{1}{21} + \frac{1}{21}}} = -1.12$$

complete 

Ex(5):

The following are the weights (kg) and blood glucose levels (mg/100 ml) of 16 apparently healthy adult males.

Weight (X)	Glucose (Y)
64.0	108
75.3	109
73.0	104
82.1	102
76.2	105
95.7	121
59.4	79
93.4	107
82.1	101
78.9	85
76.7	99
82.1	100
83.9	108
73.0	104
64.4	102
77.6	87

Given: $\sum_{i=1}^{16} x_i = 1237.8$, $\sum_{i=1}^{16} x_i^2 = 97178.6$, $\sum_{i=1}^{16} y_i = 1621$, $\sum_{i=1}^{16} y_i^2 = 165801.00$, $\sum_{i=1}^{16} x_i y_i = 126128$

- Estimate the regression line i.e. $\hat{Y} = \hat{\alpha} + \hat{\beta}X$?
- Compute the Pearson correlation coefficient (r)?
- What is the predicted glucose level for a man who weights 95 kg?

$$\bar{x} = 77.3625, \quad \bar{y} = 101.3125$$

$$S_{xx} = \sum x_i^2 - n(\bar{x})^2 = 97178.6 - 16(77.3625)^2 = 1419.2775 = S_{xx}$$

$$S_{yy} = \sum y_i^2 - n(\bar{y})^2 = 165801.1 - 16(101.3125)^2 = 1573.5375 = S_{yy}$$

$$S_{xy} = \sum x_i y_i - n \bar{x} \bar{y} = 126128 - 16(77.3625)(101.3125) = 723.3875 = S_{xy}$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{723.3875}{1419.2775} = 0.51$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} = 101.3125 - 0.51(77.3625) = 61.88$$

$$\hat{y} = 61.88 + 0.51x$$

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \frac{723.3875}{\sqrt{(1419.2775)(1573.5375)}} = 0.484$$

$$\hat{y} = 61.88 + 0.51(95) = 110.33$$

7